# Question, evidence base, and research trail

River Conwy eDNA | four-page visual brief

## 1 Our question

**What can a replicated river-eDNA dataset establish after correcting the unit of inference, testing missed-detection recovery, auditing pipeline disagreement, and challenging taxonomic and trait interpretations?**

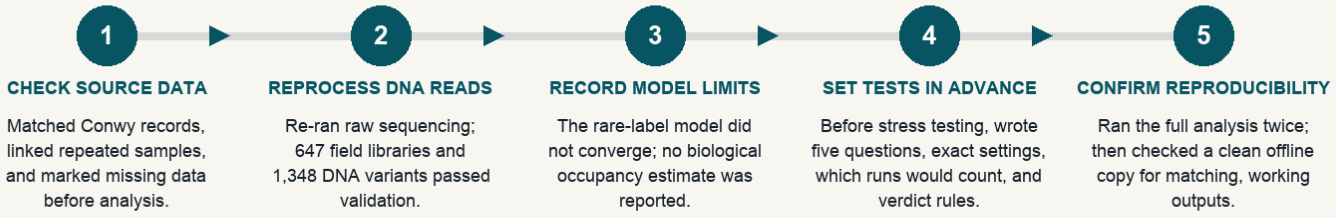
**14 sites | 18 usable timepoints | 647 validated field libraries | 1,348 ASVs | replicated 12S survey**

**Decision target: separate a stable descriptive pattern from mechanisms or corrections that the current evidence cannot identify.**

## 2 Our methodology + key research documentation moments

### How the evidence was checked and documented

Five explained steps moved the source data toward defensible claim decisions




**WHY THIS MATTERS**

The five questions and verdict rules were written before the new stress tests, so favourable results could not define the plan.

Unsuccessful models stayed visible, and running everything twice checked that the same inputs produced the same outputs.

**WHAT RE-RUNNING PROVES**

Running the workflow twice showed that the same committed inputs produced the same files and that the offline links worked. This confirms computational reproducibility; it does not add new biological replication or independent field evidence.

# Five claims enter the audit

Each hypothesis receives one decision label and one reason

## 3 Our five initial claims

| | | |
|---|---|---|
| 1 | **Detected analysis-label richness broadens downstream robustly.** | HYPOTHESIS 01 |
| 2 | **A correction model can recover true occupancy of rare Conwy labels.** | HYPOTHESIS 02 |
| 3 | **Lower source signal and processed fish counts predict pipeline non-recovery.** | HYPOTHESIS 03 |
| 4 | **Longer mitochondrial markers resolve Salmo species better than MiFish.** | HYPOTHESIS 04 |
| 5 | **Trait diversity broadens beyond what richness and prevalence alone predict.** | HYPOTHESIS 05 |

*These were hypotheses formulated by this reanalysis, not claims attributed to the source-study authors.*

## 4 Stressing those claims with more tests

### Five claims, one decision vocabulary

ACCEPTED = adequate evidence met the decision rule | REJECTED = an adequate test did not support the claim
NON-CONCLUSIVE = fit, validity, or coverage was insufficient to decide

| | | |
|---|---|---|
| **River community** | 64/72 valid (<90%); direction stable | NON-CONCLUSIVE |
| **Imperfect detection** | Real fit failed; atlas is synthetic only | NON-CONCLUSIVE |
| **Pipeline disagreement** | 999/999 finite; 43/43 leave-outs negative | ACCEPTED |
| **Taxonomic resolution** | 2 of 7 labels eligible; minimum 4 | NON-CONCLUSIVE |
| **Functional traits** | Both null p-values were > 0.05 | REJECTED |

Decision tally: 1 accepted | 1 rejected | 3 non-conclusive. Detailed decision records remain linked offline.

**Declared evidence: 64/72 community families (reference p = 0.0714); 613/1,944 recoverable occupancy cells; pipeline beta -1.149 with 999/999 finite bootstraps; 2 of 7 eligible Salmo labels; trait-null p = 0.3159 / 0.1142.**

# The modelling laboratory

MODEL 01-03 | multiverse, known-truth simulation, and cluster-aware association

## 5 The models and simulations used to evaluate these claims

A single model could not answer five different scientific questions. Each engine below matches its own estimand, dependence structure, stress test, and decision rule.

MODEL 01

### Community robustness multiverse

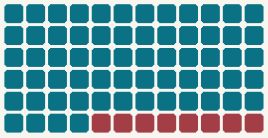
NON-CONCLUSIVE

INPUT -> MODEL FAMILY -> RESTRICTED STRESS TEST -> DECISION

**SPECIFICATION SPACE**

3 panels x 2 taxonomies
x 2 thresholds x 2 effort rules
x 3 distances = 72 families

**COMPLETE FAMILY GRID**



64 valid 8 undefined rCLR

Invalid families remain in the reported total

**RESTRICTED INFERENCE**

Site x timepoint units stay clustered
28 river arrangements: p = 2/28
3,600 draws + 75 leave-outs

**DECISION RULE**

64/72 valid = 88.9% (<90%)
64/64 positive; 3,600/3,600
0/75 sign reversals

**NON-CONCLUSIVE: 64/72 valid missed the 90% rule, although every valid direction was positive.**

MODEL 01 | NON-CONCLUSIVE | Reason: 64/72 valid (<90%); direction stable

OPEN MODEL + EVIDENCE >

MODEL 02

### Known-truth occupancy recoverability atlas

NON-CONCLUSIVE

SYNTHETIC GENERATOR -> DETECTION HISTORIES -> THREE ESTIMATORS -> DECISION MAP

**648 SCENARIOS**

psi, prevalence and baseline p
1-3 replicates + depth structure
missingness + missing-data coding

**KNOWN TRUTH**

120 ordered site x time units
latent occupancy z_i
replicate detections + missing-data rules



**ESTIMATOR LANES**

NAIVE OCCURRENCE

REPLICATION-AWARE

DEPTH-AWARE

**1,944 SCENARIO CELLS**



613 recoverable

732 unreliable

599 non-identifiable

No baseline p = 0.20 cell passed

**NON-CONCLUSIVE: the real fit failed; the 64,800 known-truth datasets provide synthetic design evidence only.**

MODEL 02 | NON-CONCLUSIVE | Reason: real fit failed; atlas is synthetic

OPEN MODEL + EVIDENCE >

MODEL 03

### Cluster-aware pipeline-disagreement model

ACCEPTED

VERIFIED CELLS -> LOGISTIC ASSOCIATION -> PAIRED CLUSTER BOOTSTRAP -> DECISION

**COMPARATOR CELLS**

2,344 source-positive
1,962 recovered
382 non-recovered

**LOGISTIC MODEL**

logit P(non-recovery)
source signal + count context
section + season

Sensitivity excludes focal label + Unassigned

**999 PAIRED BOOTSTRAPS**




43 leave-outs: 11 taxa + 14 sites + 18 times

**FOCAL-EXCLUDED RESULT**



beta = -1.149

95%: -3.086 to -0.579

999/999 finite; 43/43 negative

**ACCEPTED: 999/999 finite and 43/43 negative; comparator-relative and noncausal, not a biological false-negative rate.**

MODEL 03 | ACCEPTED | Reason: 999/999 finite; 43/43 leave-outs negative

OPEN MODEL + EVIDENCE >

# What the evidence decided - and what I added

MODEL 04-05 | the same three-state decision rule, applied to identity and traits

## 5 Models continued: identity and mechanism

MODEL 04

### Multi-marker Salmo resolution benchmark

NON-CONCLUSIVE

ACCESSIONS -> MARKER WINDOWS -> DISTANCE / BARCODE / ALL-TIES LOO -> DECISION

**PUBLIC PANEL**
21 mitogenome records
7 candidate Salmo labels
2 outgroup genera

**MARKER RULERS**
MiFish 12S 170 bp
COI 648 bp
cyt-b 1,140 bp

**CONSERVATIVE AUDIT**
normalised edit distance
exact-haplotype sharing + gaps
all tied nearest neighbours

**TWO DECISION CHECKS**
SPECIES: 2 of 7 eligible
minimum 4 -> NON-CONCLUSIVE
GENUS: 10/10 x 3
Salmo nearer than each outgroup

NON-CONCLUSIVE: only 2 of 7 labels were eligible; submitted labels remain annotations, not local species identity.

MODEL 04 | NON-CONCLUSIVE | Reason: 2 of 7 labels eligible; minimum 4

OPEN MODEL + EVIDENCE >

MODEL 05

### Fixed-margin functional-trait null model

REJECTED

OCCURRENCE + TRAITS -> GOWER -> RAO Q -> MARGIN-PRESERVING NULL -> DECISION

**154 UNIT MATRICES**
11 timepoint blocks

**GOWER**
equal-weight axes
flow + trophic
+ movement traits

**RAO Q**
unit diversity
site means
lower - upper

**FIXED MARGINS**
2 x 2 checkerboard swaps
richness + prevalence fixed
9,999 draws x 2 panels

**EMPIRICAL NULLS**
PRIMARY p = 0.3159
STRICT p = 0.1142

REJECTED: both p-values exceeded 0.05; no evidence of trait broadening beyond richness and prevalence.

MODEL 05 | REJECTED | Reason: p = 0.3159 / 0.1142; both > 0.05

OPEN MODEL + EVIDENCE >

## 6 Our final verdict

ACCEPTED = rule met with adequate evidence | REJECTED = adequate test did not support | NON-CONCLUSIVE = insufficient to decide

| | | |
|---|---|---|
| **River community** | NON-CONCLUSIVE | Reason: 64/72 valid (<90%); all 64 valid directions positive |
| **Real occupancy** | NON-CONCLUSIVE | Reason: Real fit failed; recovery atlas is synthetic only |
| **Pipeline association** | ACCEPTED | Reason: 999/999 finite; all 43 leave-outs negative |
| **Species resolution** | NON-CONCLUSIVE | Reason: Only 2 of 7 labels eligible; minimum was 4 |
| **Trait broadening** | REJECTED | Reason: p=0.3159 and 0.1142; both above 0.05 |

**Scientific addition: a reproducible credibility-audit framework whose questions, tests, and verdict rules were written in a dated project plan before analysis, showing which claims are accepted, rejected, or non-conclusive - and why.**